

Avanish Didwania

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EDUCATION

JIS University

Bachelor of Technology in Computer Science and Engineering

- Current CGPA: 8.73/10

West Bengal, India

2024 – 2028

EXPERIENCE

Gen AI Intern – Naviga Technology Solutions

May 2026 – Present

Content Credibility & Fact Verification

Remote

- Built **NoLie** ([GitHub](#)), a Chrome extension for AI content credibility and fact verification, achieving **90% accuracy** on a 50-claim evaluation set via context-aware and hallucination cross-verification.
- Built an **agentic verification mode**: a LangGraph agent (hosted on Railway) that autonomously classifies claims, checks source credibility, and runs web search (Tavily), protected by a **circuit breaker** that falls back to standard verification after repeated failures.
- Built real-time audio fact-checking (Deepgram) and a self-learning RAG cache (Supabase pgvector) with confidence-gated storage to prevent cache poisoning.
- Integrated Google Gemini multimodal API for AI-image detection and a source credibility scorer using the MBFC dataset (3,920+ domains).

Open Source Contributor – Submitty

May 2026

Rensselaer Polytechnic Institute (RPI)

Remote

- Fixed a Submitty forum notification bug via the `previewText()` helper, truncating cluttered email/homepage alerts for readability.
- Submitted PR #12868 to the official Submitty repository, an LMS used by universities worldwide.

PROJECTS

LunarAlign – Illumination-Invariant Image Registration for Chandrayaan-2 | *OpenCV, PyTorch, Kornia*

2026

- [GitHub Repository](#) – built for **SIH 2026, Problem Statement 26166 (ISRO)**
- Built a multi-sensor image registration pipeline (Phase Congruency, SuperPoint+LightGlue, LoFTR, MAGSAC++) achieving **sub-pixel accuracy (RMSE 0.16–0.21px)** on real Chandrayaan-2 TMC imagery, validated against known ground-truth warps.
- Diagnosed a 1.65px residual on a real Fore/Nadir stereo pair as **parallax-limited** (a geometric property of the stereo pair, not the algorithm), backed by per-point residual analysis showing 39% of inliers already sub-pixel.
- Engineered a coarse-to-fine two-stage refinement to bridge a 40× scale gap across sensors, and solved real production issues: 800MP OOM crashes (tiled processing), PDS4/PDS3 format mismatches (custom parser), and non-overlapping orbit strips.

DocMind – AI Document Intelligence Platform | *FastAPI, LangGraph, Auth0, PGVector*

2026

- [GitHub Repository](#)
- Built a **skills-based platform** with an LLM intent router dispatching queries to Document Q&A, Fact-Checking (calling the NoLie agent as a remote service), or General Chat.
- Implemented **self-correcting retrieval**: an LLM grades retrieved chunks and reformulates the query (up to 2 attempts) on failure, layered on hybrid retrieval (BM25 + vector + RRF).
- Secured the platform with **Auth0 (OIDC + PKCE, MFA, RBAC)**, a 3-stage security pipeline, and automated LLM-as-judge evaluation for retrieval relevance and faithfulness.

Stock Portfolio & Market Analytics Dashboard | *React.js, Node.js, Express.js, MongoDB, JWT*

2026

- [Live Demo](#) | [GitHub Repository](#)
- Built a full-stack MERN financial dashboard (JWT/BCrypt auth) with real-time portfolio tracking, integrating Alpha Vantage APIs with caching and fallback handling.

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, SQL, C

AI / ML: LLM APIs (Groq, Gemini), RAG, LangGraph, Computer Vision (OpenCV, PyTorch, Kornia), Prompt Engineering

Frontend: React.js, TypeScript, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, Chrome Extension (MV3)

Database: MongoDB, PostgreSQL, PGVector, Supabase

Tools & Auth: Git, Docker, Auth0, Postman, VS Code, Vercel

Core Subjects: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP